DIODE CIRCUITS

SEMICONDUCTOR DIODES

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

DC Input

- OR Gate and AND Gate

Sinusoidal Input

- Half-Wave Rectifier
- Full-Wave Center-Tapped Rectifier
- Full-Wave Bridge-Type Rectifier

DC INPUT

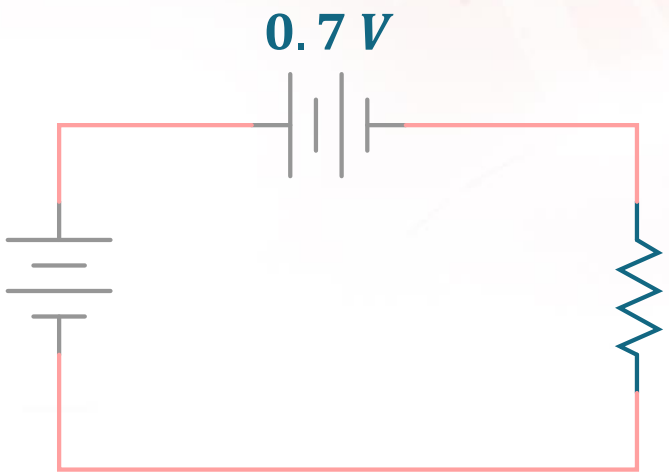
DIODE CIRCUITS

DIODE STATES

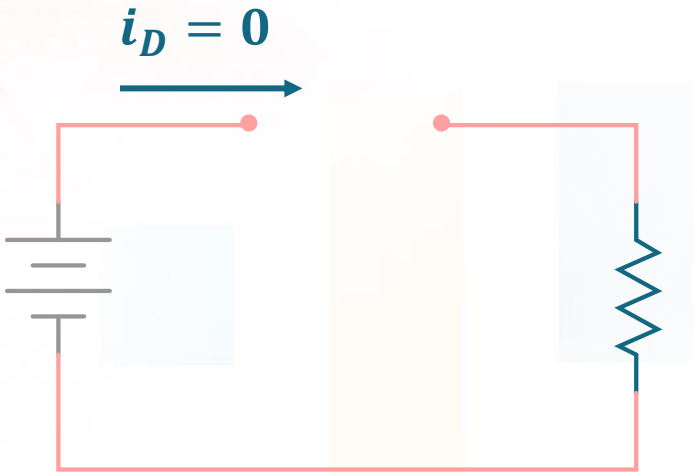
Diode is **ON**

Diode is **OFF**

Forward Bias Equivalent Circuit



Reverse Bias Equivalent Circuit



Determine the load voltage, v_L and load current, i_L for the given network.

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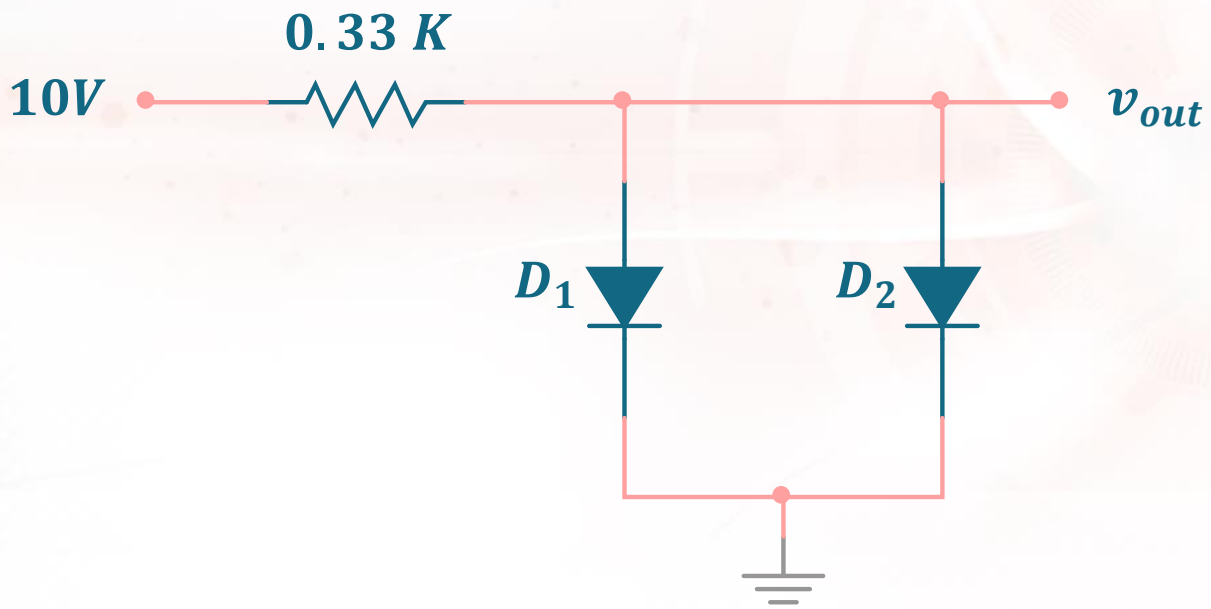
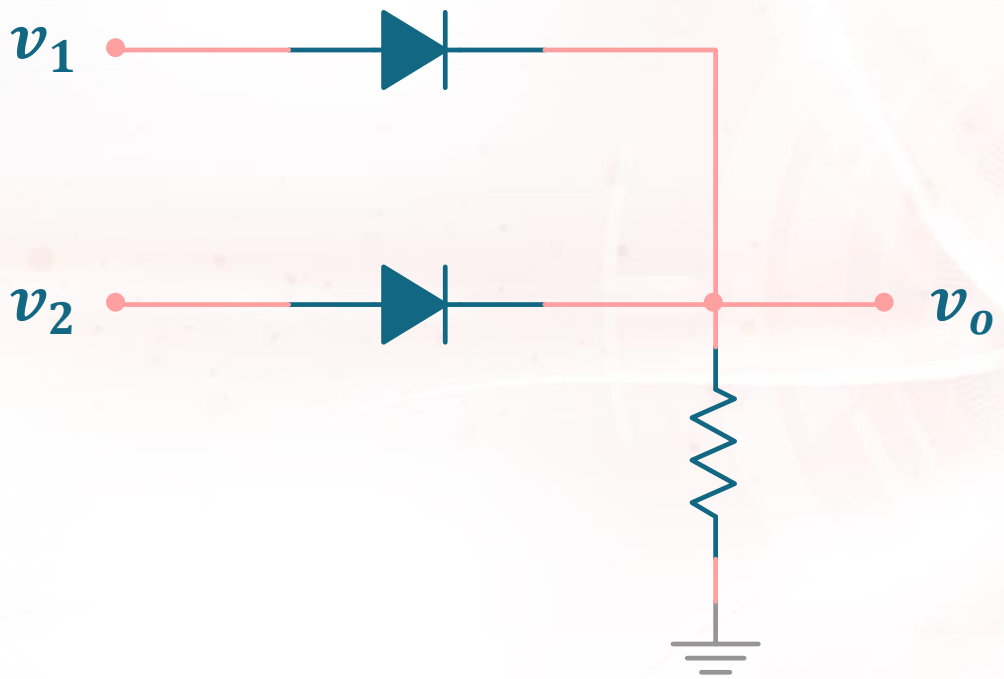
Determine the output voltage v_{out}, resistor current i_R, and diodes current i_{D1} and i_{D2} respectively for the given network.

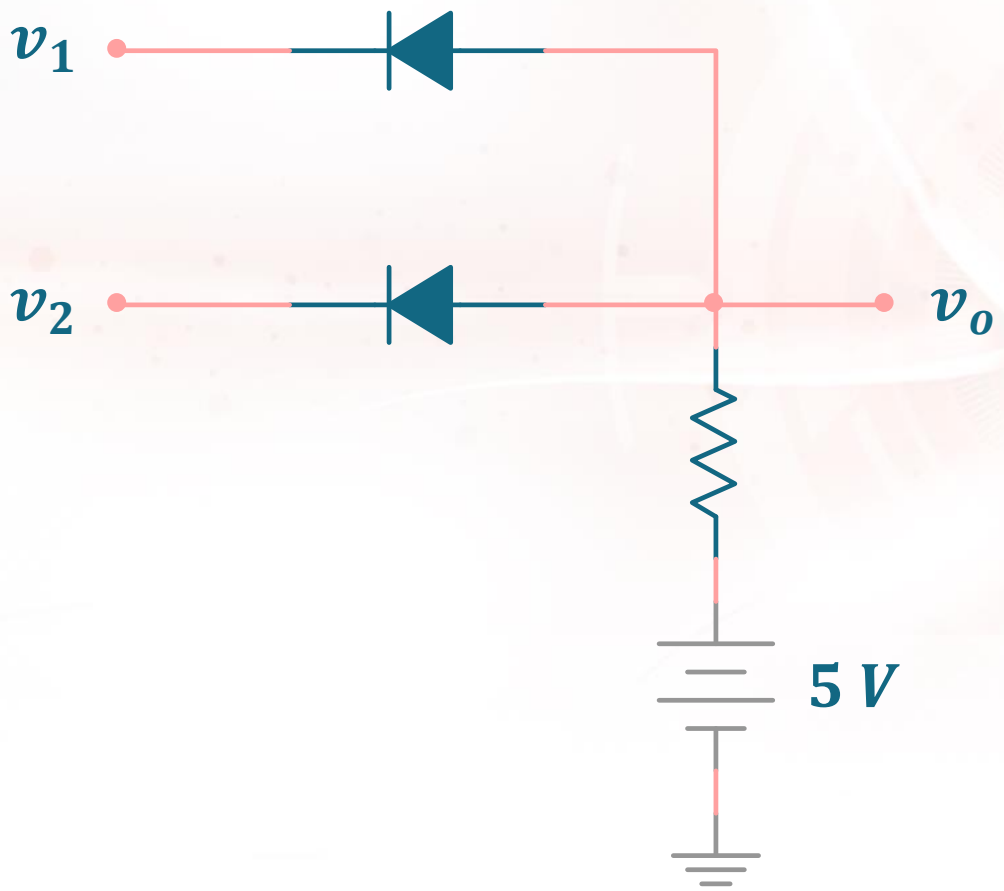


POSITIVE LOGIC OR GATE



v_1	v_2	v_o
0	0	
0	5	
5	0	
5	5	

POSITIVE LOGIC AND GATE

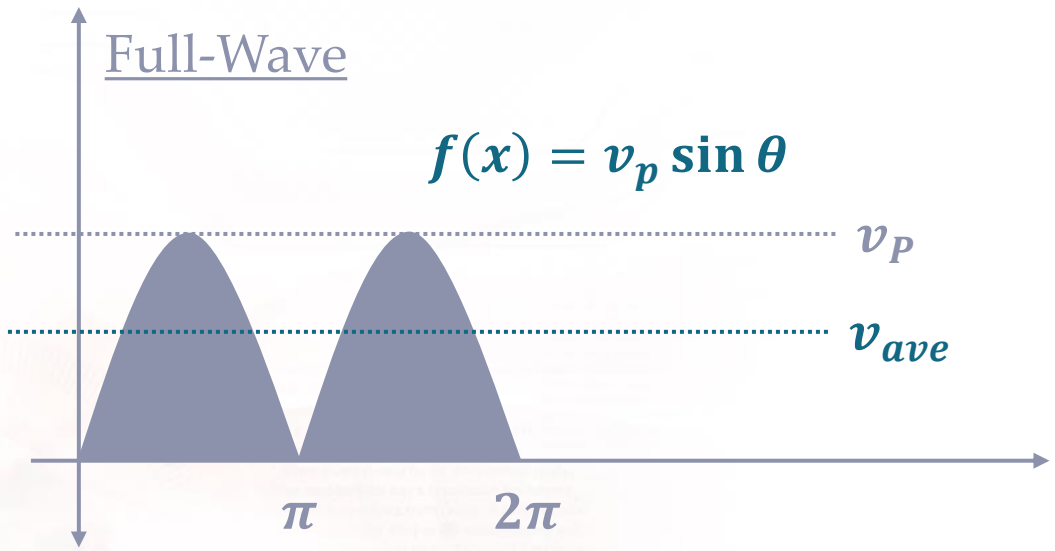


v_1	v_2	v_o
0	0	
0	5	
5	0	
5	5	

SINUSOIDAL INPUT DIODE CIRCUITS

AVERAGE VALUE OF $f(x)$

$$f(x)_{ave} = \frac{1}{b-a}\int_a^b f(x)dx$$



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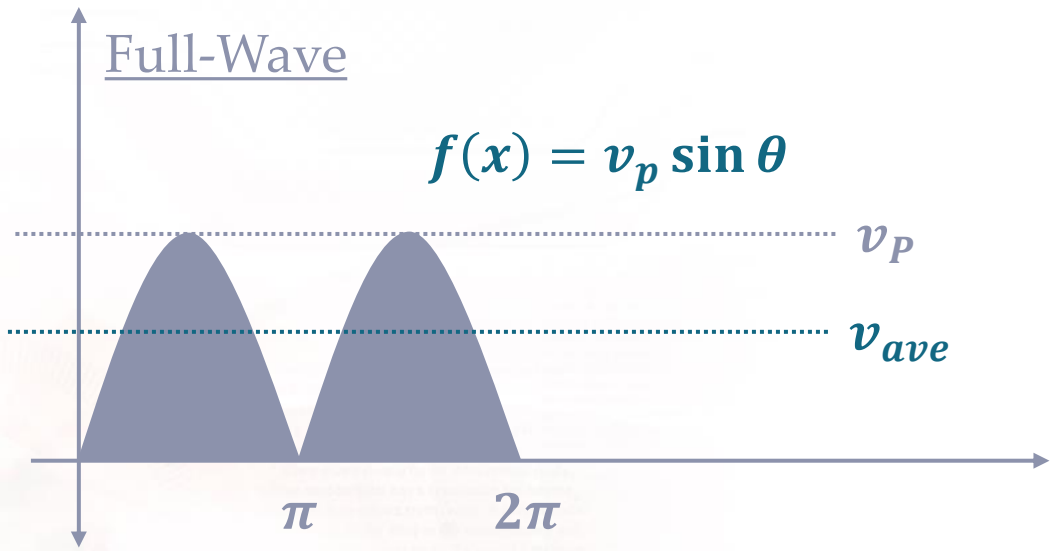

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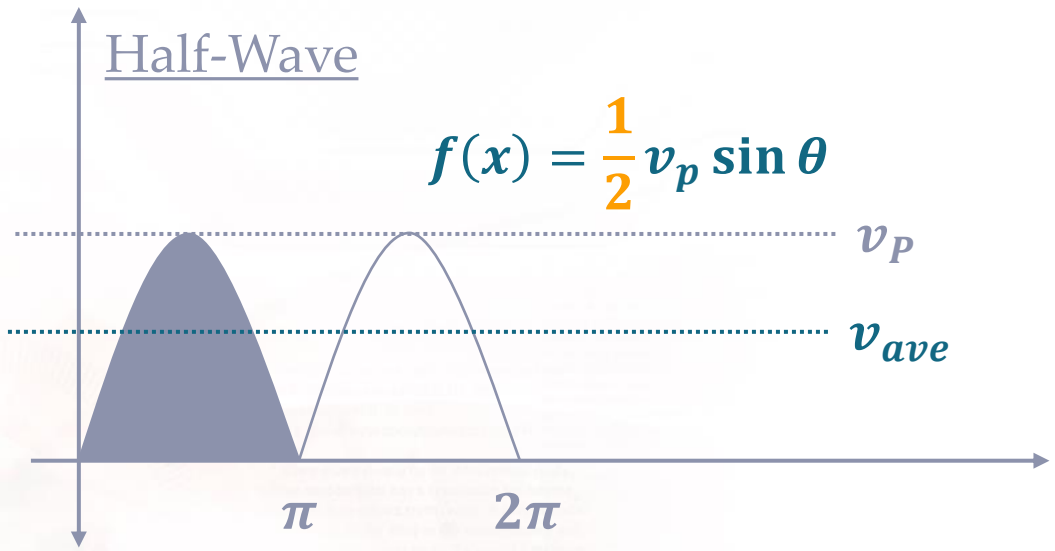
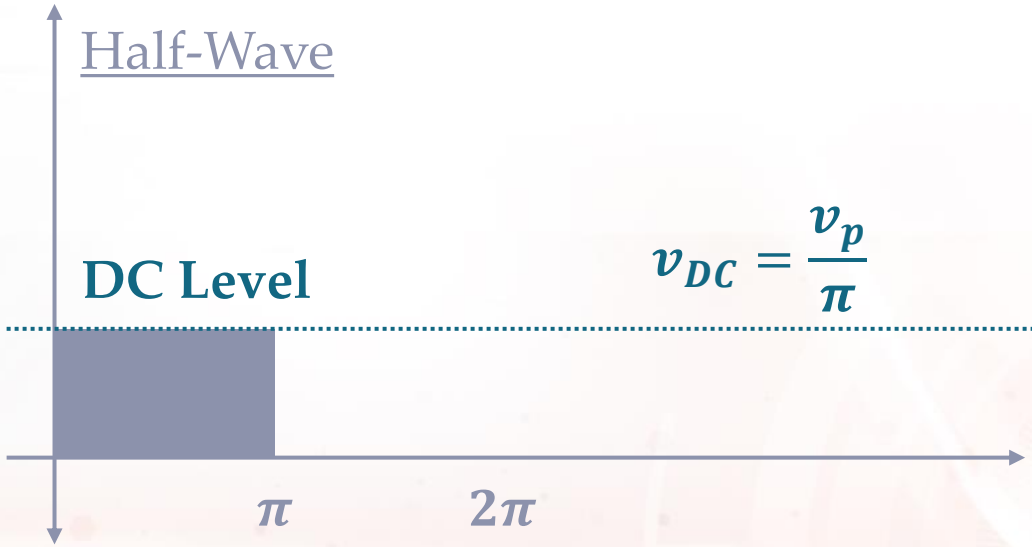
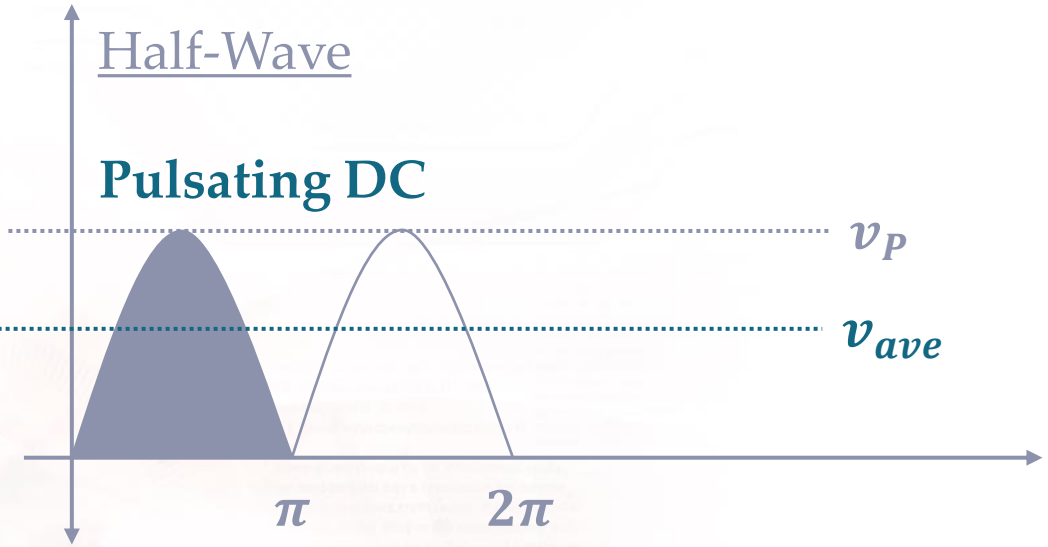
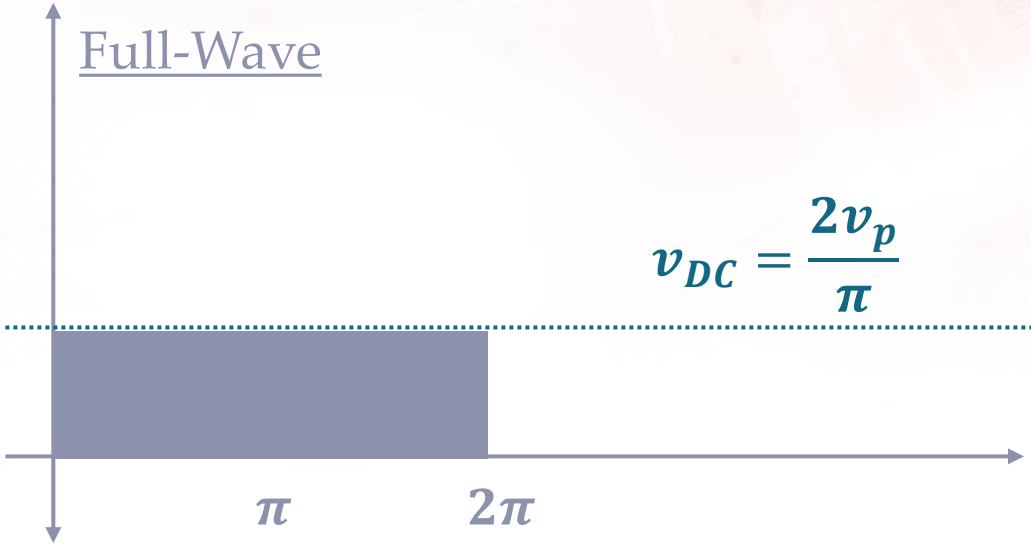
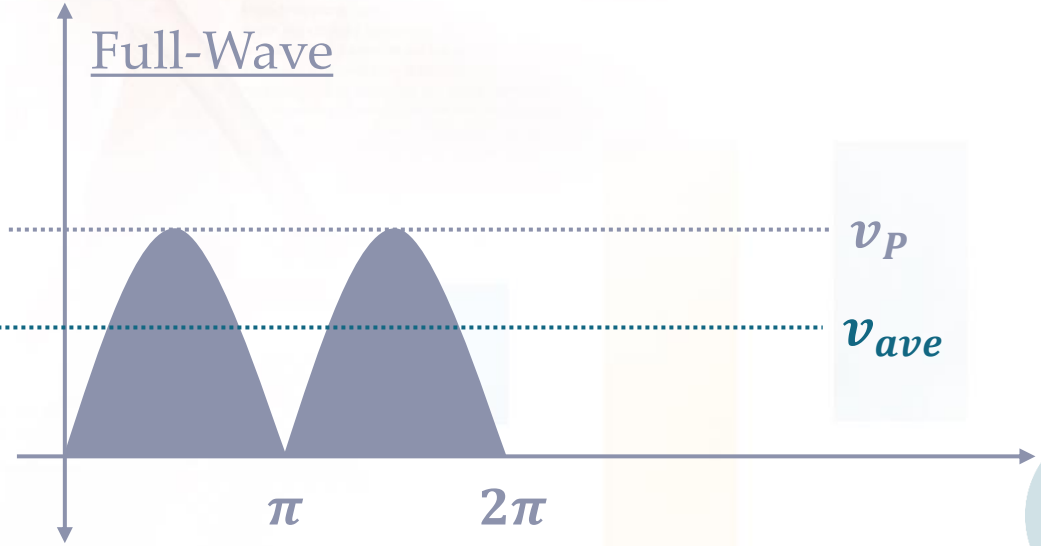


DC LEVEL

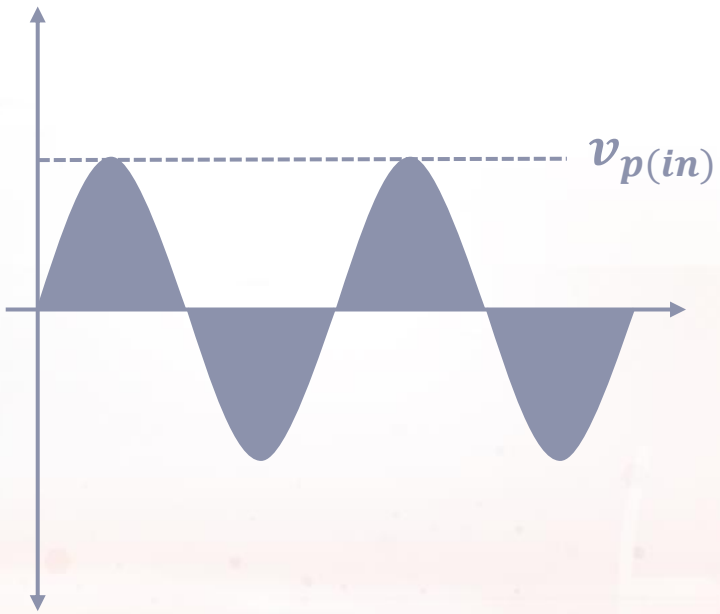

Half-Wave
DC Level
$v_{DC} = \frac{v_p}{\pi}$
π
2π

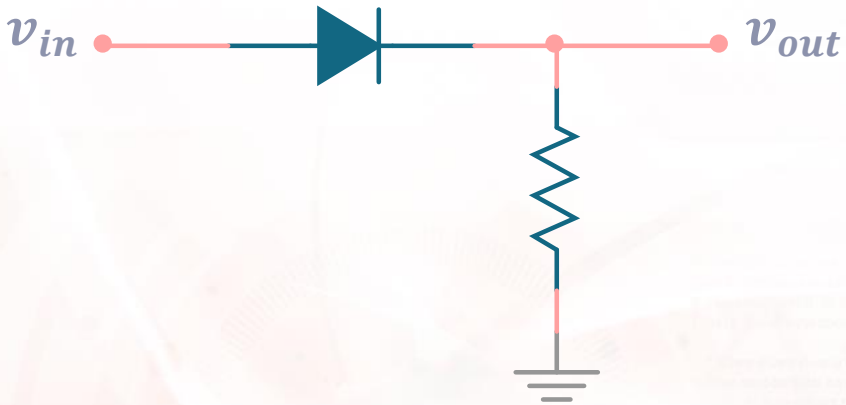

Half-Wave
Pulsating DC
v_P
v_{ave}
π
2π

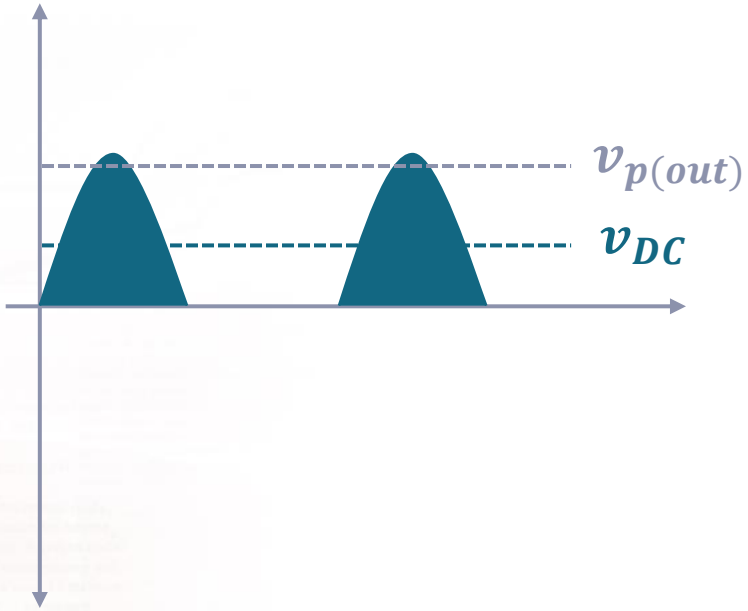

Full-Wave
$v_{DC} = \frac{2v_p}{\pi}$
π
2π


Full-Wave
v_P
v_{ave}
π
2π

HALF-WAVE RECTIFICATION







DC Level,

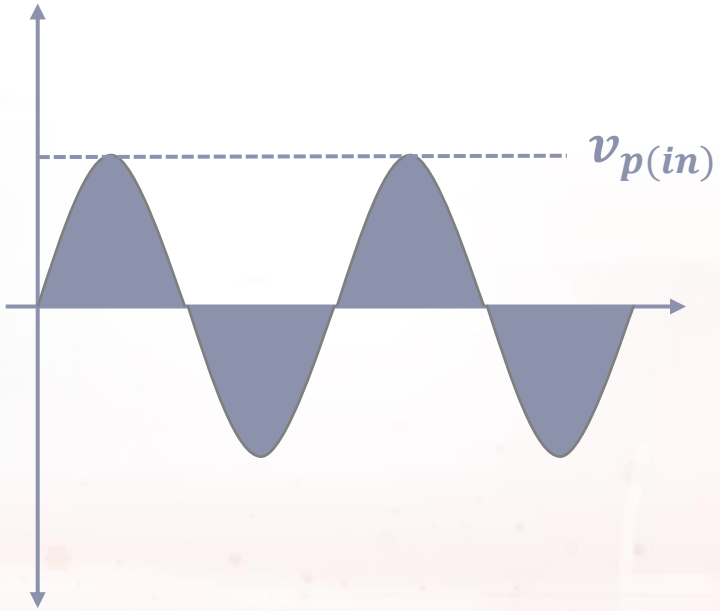
$$v_{DC} = \frac{v_{p(out)}}{\pi}$$

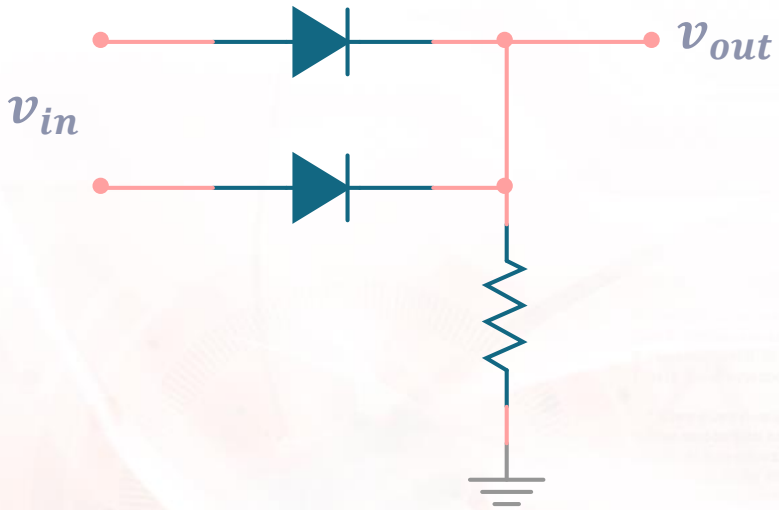
Output Frequency,

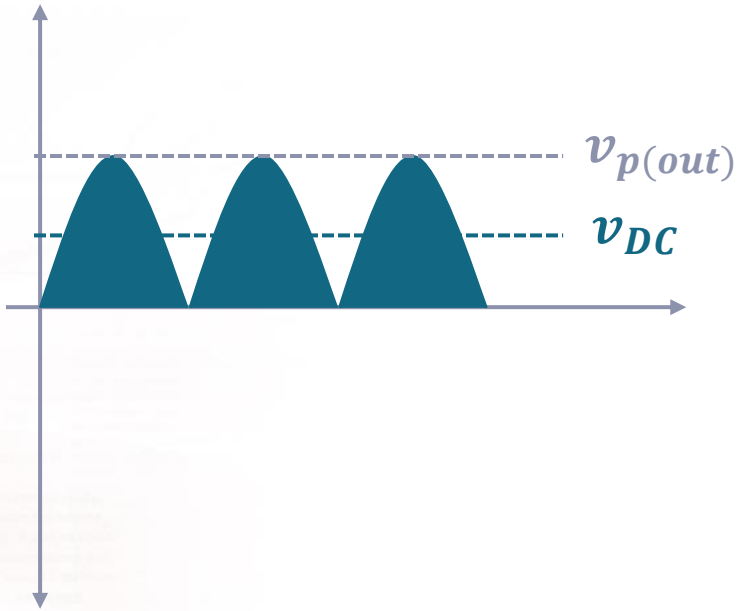
$$f_{out} = f_{in}$$

Sketch and determine the **dc level** of the output for the given network.

CENTER-TAPPED







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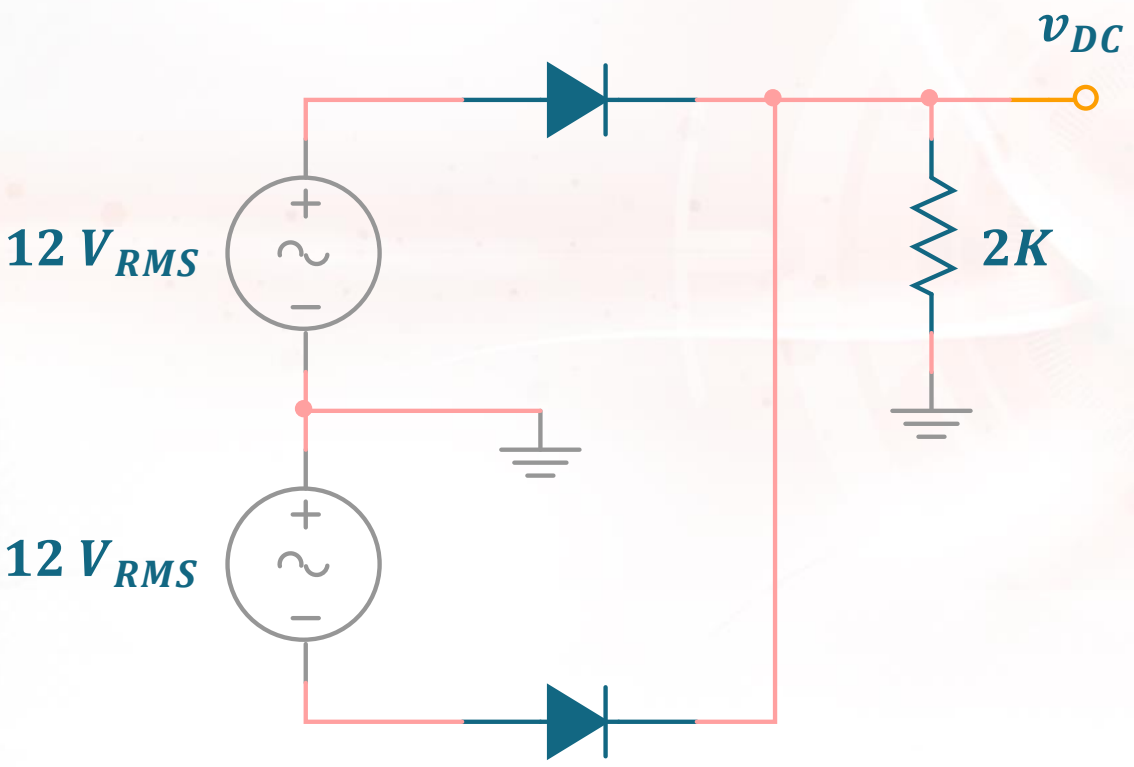
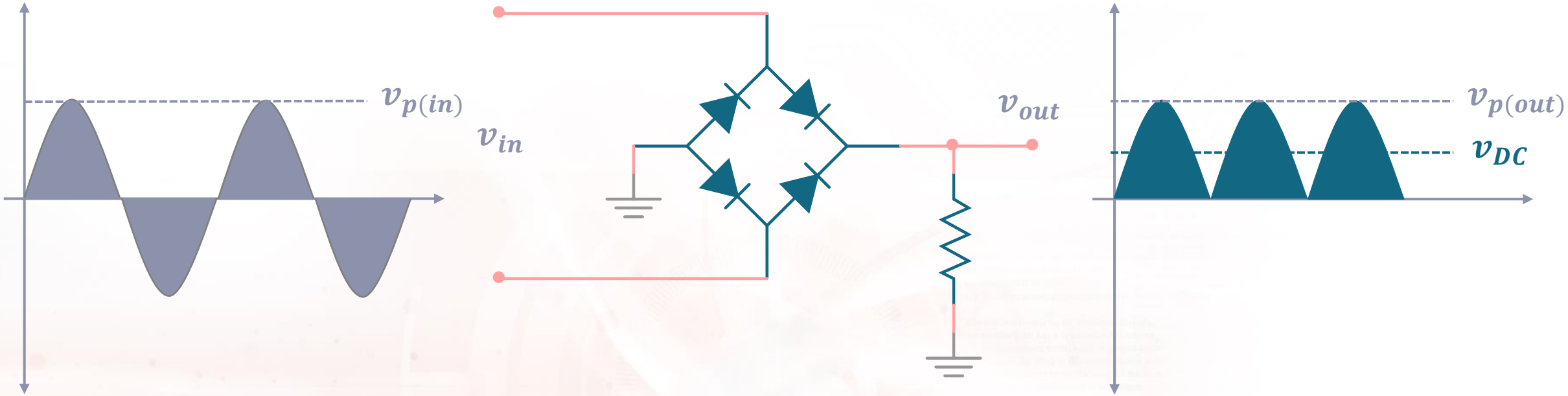


BRIDGE TYPE



DC Level,

$$v_{DC} = \frac{2v_{p(out)}}{\pi}$$

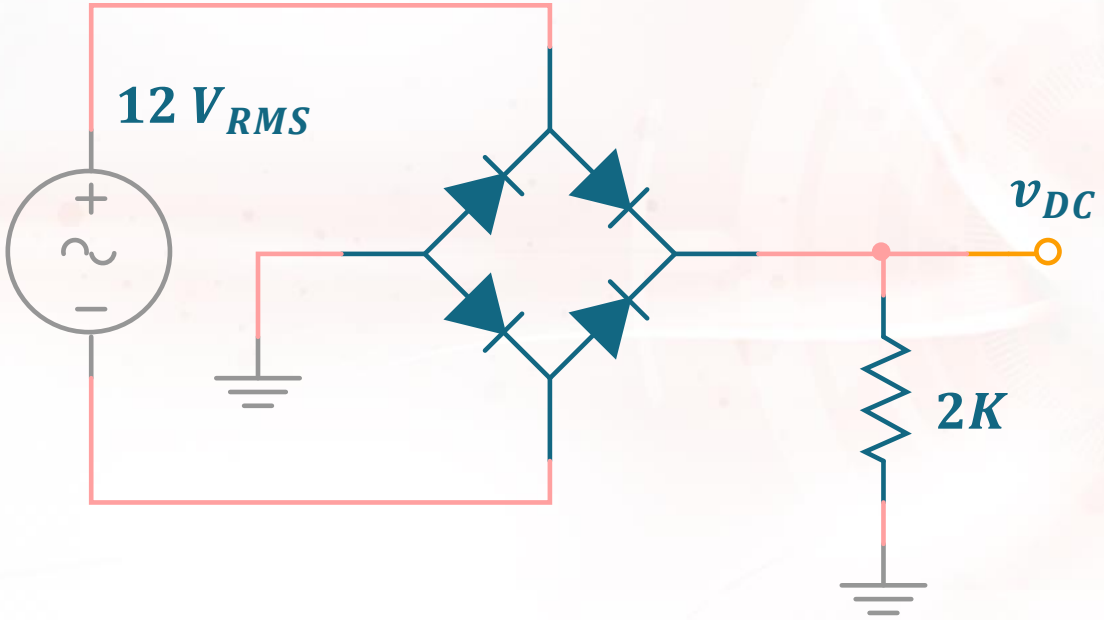
note,

two diodes are ON every half-cycle

Output Frequency,

$$f_{out} = 2f_{in}$$

Sketch and determine the **dc level** of the output for the given network.



LABORATORY